

FLASH DRUM VER 1

Operator Training Simulator · D-400 with E-400 feed preheater

Chulalongkorn University · Department of Chemical Engineering

DOCUMENT NO. : FD-V1-IN-CE-001

DOCUMENT TITLE : CAUSE AND EFFECT DIAGRAM

DRAWINGS : FD-400-PID-001 / FD-400-PID-002

Teaching document generated from the simulator logic (bmsTrip, bmsStep, tripConditions, alarm list).

Rev.	Date	Revision status	Prepared	Checked	Approved	Remark
R0	06-Oct-26	Issued for teaching	A. Soottitantawat	—	—	

FLASH DRUM VER 1 · D-400 CAUSE and EFFECT DIAGRAM Chulalongkorn University · Department of Chemical Engineering								Client :		Chulalongkorn University · Department of Chemical Engineering					REV	DATE	ORIGINATOR	CHECKER	APPROVER
								PROJECT :		Flash Drum Ver 1 · OTS					R0	06-Oct-26	A. Soottitantawat	—	—
								DOCUMENT No. :		FD-V1-IN-CE-001									
								DOCUMENT TITLE :		Cause and Effect Diagram									
								P&ID :		FD-400-PID-001 / FD-400-PID-002									SHEET 1 of 1
Causes								EFFECTS											
								XV-408 · Feed shut-off valve (fail closed)	XV-409 · Hot-oil shut-off valve to E-400 (fail closed)	XV-418 · Bottoms shut-off valve (fail closed)	I-1 · SIS I-1 trip latch (feed and hot oil)	I-2 · SIS I-2 latch (bottoms)	1ST · First-out annunciation (I-1)	RED · SIS trip alarm on the DCS (red)	AMB · Process alarm on the DCS (amber)	AUD · Audible alarm (horn)			
Trip Action								C	C	C	TP	L	F	A	A	A	Notes: 1. Action codes: C = close; TP = I-1 tripped and latched (manual reset); L = I-2 latched (XV-418 held shut); F = first-out; A = activate; P = permissive (the valve cannot be opened until the condition clears); RB = reset blocked. 2. The SIS acts only on the three shut-off valves. The BPCS loops keep their mode: FIC-400 and TIC-401 in AUTO drive FCV-400 and TCV-401 to 100 % behind the closed XV-408 and XV-409 (surge on reset), and PIC-420 keeps holding the pressure. 3. ESD push-button HS-400: immediate, no delay; closes all three XVs and latches I-1 with first-out "Operator emergency shutdown". 4. Time delays are applied in the logic solver; I-1 latches until reset at the SIS console with level < 85 % and pressure < 230 kPa g. 5. I-2 is armed only while XV-418 is open; it does not stop the feed, so the level then climbs to LSHH-416 (HAZOP R-20). 6. Process alarm suppressed while the unit is not meant to be feeding (shutdown, start-up before feed). 7. PSV-400 is a relief device, not an SIS cause: shown for completeness of the D-400 annunciation; it starts to lift about 25 kPa below its set. 8. Sequence permissive, not a trip: no valve moves; the operator cannot open the valve or reset until the condition clears. 9. LT-400 (BPCS) is not used by any trip; LSHH-416 and LSL-417 are separate level switches and PSHH-426 a separate pressure switch (independent of PT-420).		
Item No.	Tag No.	Tag Service Description	Trip/Alarm-Direction	Setting	Eng. Unit	Delay	Note no.												
001	HS-400	Emergency shutdown push-button (DCS / SIS console)	Pushed	—	—	0 s	3	X	X	X	TP		X	X		X			
002	LSHH-416	D-400 level high-high: liquid toward the vapour line (own level switch)	High-High	90	% span	5 s	4	X	X		TP		X	X		X			
003	PSHH-426	D-400 pressure high-high (own pressure switch)	High-High	250	kPa g	3 s	4	X	X		TP		X	X		X			
004	LSLL-417	D-400 level low-low: gas blow-by to the liquid line (own level switch)	Low-Low	5	% span	5 s	5			X		L		X		X			
005	FIC-400	Feed flow low (FT-400)	Low	5	m³/h		6								X	X			
006	FIC-400	Feed flow high (FT-400)	High	20	m³/h		6								X	X			
007	TIC-401	Hot feed temperature low (TT-401)	Low	69	°C		6								X	X			
008	TIC-401	Hot feed temperature high (TT-401)	High	79	°C		6								X	X			
009	LIC-400	D-400 level low (LT-400)	Low	20	% span										X	X			
010	LIC-400	D-400 level high (LT-400)	High	80	% span										X	X			
011	PIC-420	D-400 pressure low (PT-420)	Low	70	kPa g										X	X			
012	PIC-420	D-400 pressure high (PT-420)	High	130	kPa g										X	X			
013	FI-420	Overhead vapour flow low	Low	300	Nm³/h		6								X	X			
014	FI-420	Overhead vapour flow high	High	900	Nm³/h		6								X	X			
015	FIC-410	Bottoms liquid flow low	Low	8	m³/h		6								X	X			
016	FIC-410	Bottoms liquid flow high	High	18	m³/h		6								X	X			
017	AI-410	Bottoms butane above the 28 wt% spec	High	28	wt% C4		6								X	X			
018	AI-420	Overhead butane below the 75 wt% spec	Low	75	wt% C4		6								X	X			
019	AI-400	N2 in the drum vapour high	High	5	% N2		6								X	X			
020	TI-410	Flash temperature low (drum liquid)	Low	40	°C		6								X	X			
021	TI-410	Flash temperature high (drum liquid)	High	60	°C		6								X	X			
022	PDI-406	Hot-oil header ΔP low	Low	100	kPa		6								X	X			
023	PDI-406	Hot-oil header ΔP high	High	200	kPa		6								X	X			
024	PSV-400	Relief valve lifting to flare (set at the MAWP)	Lifting	450	kPa g		7								X	X			
025	D-400	Liquid carried over into the vapour line	Carry-over	—	—										X	X			
026	SIS-RST	I-1 reset permissive: level < 85 % and pressure < 230 kPa g	Not met	85 / 230	% / kPa g		8	P	P		RB								
027	XV-418	XV-418 open permissive: level above 10 % and I-2 reset	Not met	10	% span		8			P		RB							
028																			
029																			